



CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

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CLASS

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NUMBER

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BIOLOGY

Paper 3: Applications paper and Planning Question

9648/03

29th August 2014

2 hours

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your index number and name on all the work you hand in.

Write in dark blue or black pen on both sides of the paper. **[PILOT FRIXION ERASABLE PENS ARE NOT ALLOWED]**

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

There are two sections in this paper.

Section A]

Answer all questions

Section B]

Answer all questions. Answer each part on a **separate** piece of paper.

At the end of the examination, fasten all work securely together.

The number of marks is given in brackets [] at the end of each question or part of the question.

For Examiner's Use	
Section A	52
1 [12]	
2 [14]	
3 [14]	
4 [12]	
Section B	20
5a [6]	
5b [6]	
5c [8]	
TOTAL	72

Section A

Answer **all** questions in this section.

1. **Fig. 1.1** shows the plasmid pEOD.

Plasmid pEOD is a cloning vector used in creating recombinant DNA. It contains two selectable marker genes, Kan^R (kanamycin resistance gene) and Amp^R (ampicillin resistance gene). A multiple cloning site (MCS) is located within the Kan^R gene.

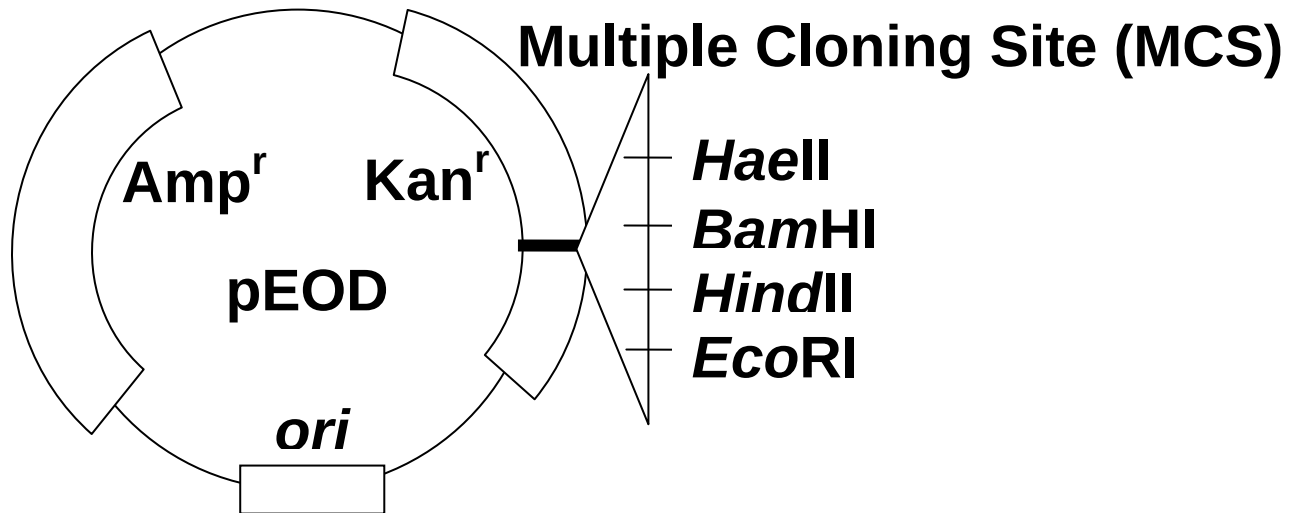


Fig. 1.1

A target sequence of human DNA is replicated using the Polymerase Chain Reaction (PCR).

The target DNA sequence is shown in Fig. 1.2.

5' - GGATCCATCCCGATCGATCCTTTAAGCTAGCTAGGATCC - 3'
3' - CCTAGGTAGGGCTAGCTAGGAAATTCGATCGATCCTAGG - 5'

Fig. 1.2

- (a) Design two 8-base long primers for the sequence to be replicated.

[2]

- (b) State three differences between the process of PCR and natural DNA replication in cells. [3]

After the target DNA has been subjected to PCR, it is cleaved using restriction enzymes and then inserted into pEOD. **Fig. 1.3** shows the types of restriction enzymes that can be used to cleave pEOD at the Multiple Cloning Site (MCS) shown in **Fig. 1.1**

Restriction enzyme	Specific target base sequence of DNA
HaeIII	<pre> 5' ... G G C C ... 3' 3' ... C C G G ... 5' </pre>
BamHI	<pre> 5' ... G G A T C C ... 3' 3' ... C C T A G G ... 5' </pre>
HindIII	<pre> 5' ... A A G C T T ... 3' 3' ... T T C G A A ... 5' </pre>
EcoRI	<pre> 5' ... G A A T T C ... 3' 3' ... C T T A A G ... 5' </pre>

Fig. 1.3

- (c)(i) State the restriction enzyme used to cleave the target DNA sequence before it is inserted into vector pEOD. [1]

(ii) Outline the steps that must subsequently be followed to produce a recombinant plasmid. [3]

(d) Describe and explain how bacteria transformed with this recombinant plasmid are identified using selective agar media. [3]

[Total: 12]

- 2 In Singapore, more than 20% of all cancers diagnosed in women are breast cancers. Mutation in BRCA-1 gene is strongly associated with the heritable form of this disease. BRCA-1 gene codes for a protein involved in cell cycle regulation and DNA repair. Fig 2.1 shows the pedigree and RFLP profile of a family with history of heritable breast cancer.

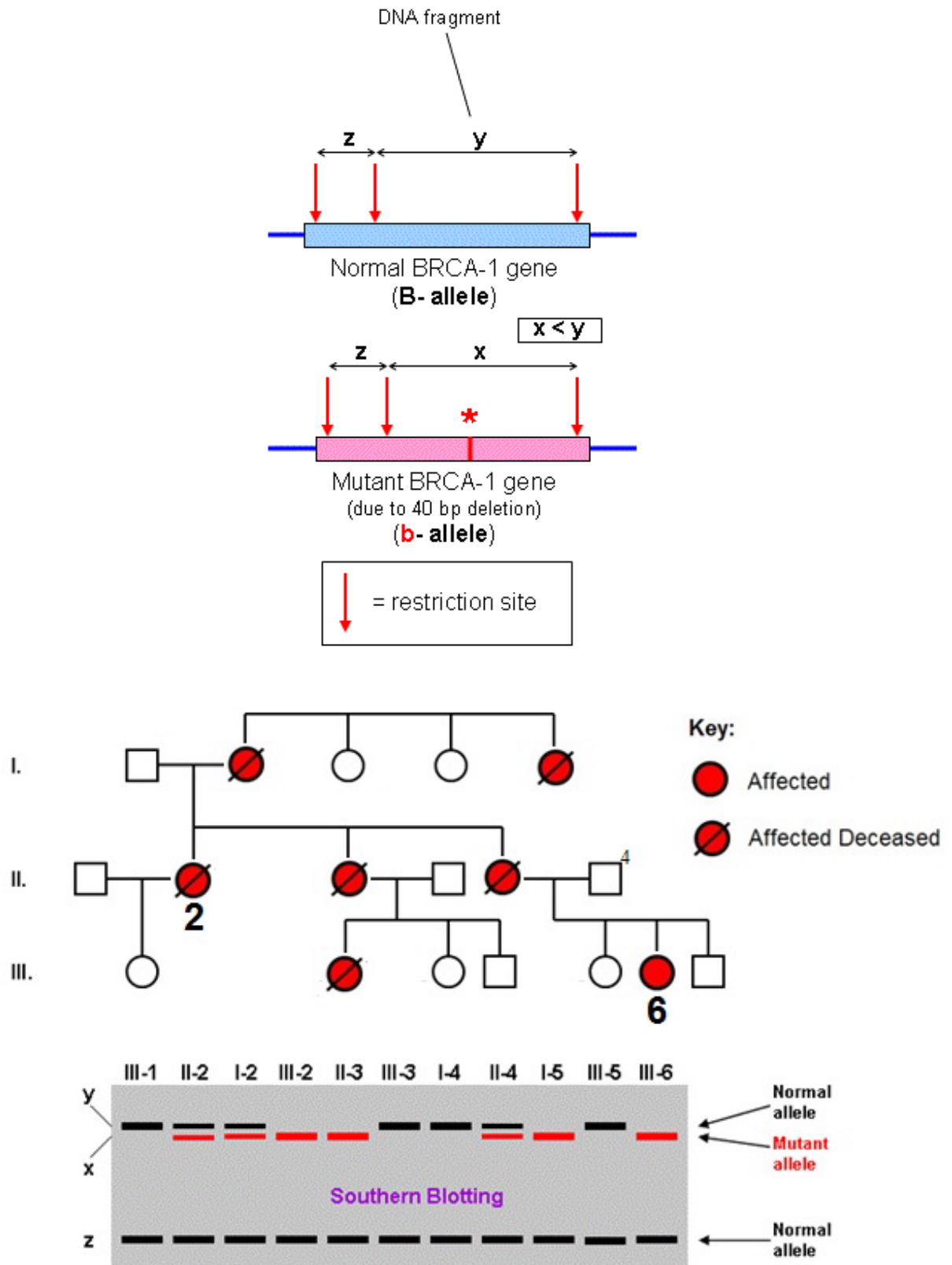


Fig 2.1

- (a)** State the genotype(s) that will lead to development of this form of breast cancer among females. [1]

- (b)** Explain how the mutation on b-allele allows for the RFLP analysis of this heritable form of breast cancer. [3]

- (c)** Account for the band patterns of the following female individuals: [4]

II-2:

III-6:

(d) Explain how bands patterns are formed in RFLP. [3]

(e) Suggest how the probe used in the RFLP analysis can be produced. [3]

[Total: 14]

- 3** The cystic fibrosis transmembrane conductance regulator (CFTR) gene encodes for the CFTR chloride channel. The CFTR chloride channel is responsible for transporting chloride ions in and out of the cell.

Individuals who suffer from cystic fibrosis (CF) have genetic mutations on the CFTR gene which results in dysregulation of chloride ions across the cell. As shown in Fig. 3.1, thick mucus produced in CF patients clogs up the airways of the lungs and the branches of the pancreatic duct.

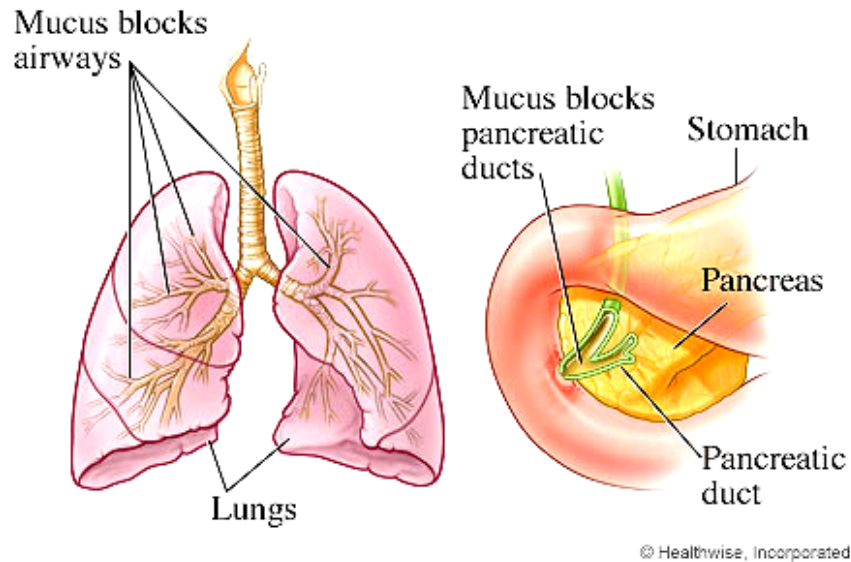


Fig. 3.1

- (a)** State two more organs not shown in Fig. 3.1 which may be affected by the cystic fibrosis disease. [2]

- (b)** Explain how a named mutation of the CFTR gene can lead to the condition shown in Fig. 3.1. [4]

(c) With the advancement of medical science, cystic fibrosis can now be treated via gene therapy methods.

(i) Describe one viral method of gene therapy treatment for cystic fibrosis. [2]

(ii) Describe one non-viral method of gene therapy treatment for cystic fibrosis. [2]

(d) (i) Suggest one advantage and one disadvantage of the viral method described in (c)(i). [2]

(ii) Suggest one advantage and one disadvantage of the non-viral method described in (c)(ii). [2]

[Total: 14]

4 Planning Question

A mutant strain of *Escherichia coli* (*E.coli*) has a disabled lac operon. The enzyme β -galactosidase, which is encoded by *lacZ* of the operon, has part of the polypeptide chain missing and is inactive. This mutant strain is referred to as **lac⁻**.

In genetic engineering, a modified F plasmid can be used as a cloning vector. This modified plasmid has part of the lac operon (*lacZ* gene) and is able to code for functional β -galactosidase. An active enzyme is synthesized if the plasmid is transferred to the **lac⁻** strain of *E.coli*.

Fig. 4.1 shows a simplified diagram of this plasmid.

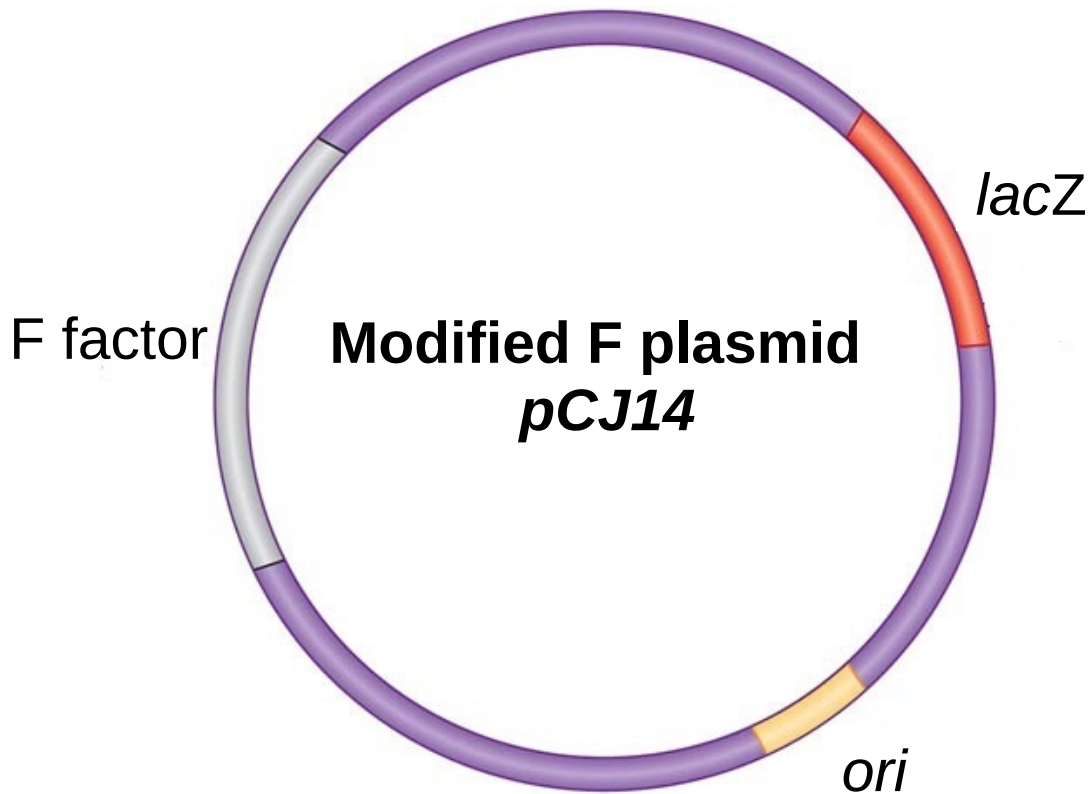


Fig. 4.1

The presence of the active enzyme β -galactosidase in colonies of **lac⁻** *E.coli* containing the modified F plasmid (**F⁺**) can be detected using nutrient agar containing the substrate, X-gal. The active enzyme hydrolyses X-gal to form a blue coloured compound. Without this plasmid the **lac⁻** colonies are white.

You are required to plan an experiment to investigate if the **efficiency of conjugation is dependent on the duration of mating time**. The transfer of the modified F plasmid into the **lac⁻** strain of *E. coli* from a **his⁻** strain of *E. coli* containing the modified plasmid (**F⁺**). The **his⁻** strain of *E. coli* lacks the genes to synthesise the amino acid histidine. It can therefore grow **only** on nutrient media to which histidine has been added.

At least 15 minutes of mating time is required for conjugation.

Conjugation efficiency is calculated as follows:

(Average number of blue colonies/Average number of blue and white colonies) x 100%

Warning: X-gal growth medium is an irritant to eyes, skin and the respiratory system.

Your planning must be based on the assumption that you have been provided with the following equipment and materials:

- Two cultures of the bacterium *Escherichia coli* (*E.coli*) are provided:

Culture A (8 ml): F^- , lac^- , his^+ *E.coli*,

Culture B (8 ml): F^+ , lac^+ , his^- *E.coli*.

- Growth medium (agar) without the amino acid histidine, X-gal provided separately
- Petri dishes
- Boiling tubes
- Sterile cotton wool plug (for boiling tube)
- Sterile graduated Pasteur pipettes
- Stopwatch
- Sterile spreader
- Disinfectant solution
- Bunsen flame and lighter
- Incubator set at 37°C
- Masking tapes
- Disposable gloves
- Protective goggles

Your plan should have a clear and helpful structure to include:

- A description of the method used including the scientific reasoning behind the method,
- An explanation of the dependent and independent variables involved, relevant, clearly labeled diagrams,
- How you will record your results and ensure they are as accurate and reliable as possible,
- Proposed layout of results tables and graphs with clear headings and labels,
- The correct use of technical and scientific terms,
- Relevant risks and precautions taken

[Total: 12]

[illegible]

[illegible]

[illegible]

Section B

Answer **all** questions. Answer each part on a **separate** piece of paper.

Write your answers on separate answer paper provided.

Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answer must be in continuous prose, where appropriate.

Your answer must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 5 (a)** Outline the procedure for propagating plants by tissue culture technique. [6]
- (b)** Describe with an example the genetic engineering methodology which utilises bacteria that can improve the nutritional content of crops [6]
- (c)** Discuss the pros and cons of genetically-modified crop plants. [8]

[Total: 20]

END OF PAPER

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